

Structurally Valid, Semantically Wrong: Measuring the Validation Gap for AI-Generated Web Inputs

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Abstract—A web application’s schema validation checks that an input is well formed. It does not check the business rules that decide whether the input is appropriate, and in the application studied here most of those rules are not enforced in code. This paper measures how often inputs that pass validation still break such rules, and whether inputs written by large language models (LLMs) differ from inputs produced in other ways. A Next.js testbed with an enumerated reference of 43 business rules received 3881 inputs through its real HTTP validation path. The inputs came from seven generation conditions: two LLM conditions, random values, rule-based adversarial values, boundary values, and two reference conditions. Semantic validity was decided by rule functions, by an independent judge model and by a human annotation of 200 inputs. Of the decided, structurally valid LLM inputs, 30.9 % (95 % CI 28.2–33.7) broke at least one rule. The rate was 20.3 % when the model was asked only for legitimate values. The gap is not specific to AI. Random structurally valid inputs broke rules more often (53.3 %), and giving the model the field’s purpose and its rules lowered the odds of a violation by 77 %. Five candidate validation layers were then evaluated. None reached the pre-registered target of 70 % recall at a 5 % false-positive rate. The best fast design detected 27.1 %, and most of what it caught were violations of rules the application could enforce in code. On a four-core CPU the rule layer added at most 14 ms to p97.5 latency and the embedding layer 65–96 ms for a single client. At the median the small language model added about half a second and the LLM judge 1.6–7.7 s, and the judge saturated on two of three endpoints at eight concurrent clients. The study covers one application and CPU-only open models. The judge agreed with a human annotator at Cohen’s $\kappa = 0.24$, so results that depend on judgement rules are exploratory.

Index Terms—input validation, business logic, large language models, web applications, software testing, empirical study

I. INTRODUCTION

A web application accepts an input only after it passes validation. In the JavaScript and TypeScript ecosystem that validation is usually a schema. The schema is a declarative object that states the type, presence, length, pattern, numeric range, enumeration and format of every field. It is checked on the client, again on the server, and finally by database constraints. Schemas of this kind are well understood [1], [2] and have been studied empirically at scale [3]. Other rules depend on what the application is for: a promotional code has to apply to the item being bought, a workshop cannot start in the past, and a review is supposed to be about the workshop it is attached to. Some of these can be written as custom refinements, although in practice most are never written down

anywhere. The distinction that matters is therefore between the business rules an application enforces and the many it leaves unenforced.

That limitation is not new. Felmetzger et al. separated input-validation flaws, which admit a general specification, from logic vulnerabilities, which do not [4]. Later work detected such flaws from behavioural traces [5], from client-server validation mismatches [6] and from over-permissive API payloads [7]. What has changed since then is who writes the input. Large language models (LLMs) now fill forms, operate web applications on a user’s behalf, and generate test data [8]–[10].

Recent work shows that LLM output which satisfies a schema is often wrong in meaning. Schema-constrained ordering agents reach full schema validity while semantic success falls as low as a few per cent [11]; hard schema decoding raised validity while executable accuracy fell [12]; across 21 models, schema compliance exceeds value accuracy by 15 to 25 percentage points [13]; schema-first tool interfaces increase schema-valid semantic misuse [14]; and in one enterprise platform the most structurally successful model had among the lowest semantic satisfaction [15]. All of this concerns model outputs that a tool, an API or a renderer consumes, and there a wrong value is discarded. At the input boundary of a web application the accepted value is stored and later shown to other users or to an assistant, and two questions remain open there. How often do LLM-written inputs that pass validation break the application’s business rules? And is that rate different from the rate for other producers of structurally valid input? Web testing studies cannot answer either, because they treat acceptance by the application as validity [8], [9]; one of them notes that incomplete e-mail addresses may be submitted successfully because the application lacks the check [8], and a robustness study using status codes as its oracle cannot see failures that return success [16]. Where semantic validity is measured, a base eight-billion-parameter model produces domain-coherent values under a quarter of the time [17], with no non-LLM comparison.

This paper measures the gap at the input boundary of one purpose-built application that has an explicit rule reference. It compares LLM inputs with random, rule-based, boundary and reference inputs on the same outcome. The contributions are:

- **C1.** A measurement of how often structurally valid inputs break unenforced business rules, for LLM and non-LLM

producers alike, broken down by prompt intent, field category and target, with a sensitivity analysis for every design choice that affects the rate.

- **C2.** A controlled comparison of seven generation conditions on one outcome, showing that the gap is not specific to AI-generated input: random structurally valid inputs exceed LLM inputs, and field context reduces LLM violations.
- **C3.** An evaluation of five validation layers (rules, embedding similarity, a small language model, a hybrid and an LLM judge) at a bounded false-positive rate on held-out fields, none of which reaches the pre-registered target.
- **C4.** Their request-path cost on commodity CPU hardware, a regime for which the guardrail literature I reviewed reports no comparable figures [18]–[21].
- **C5.** The testbed, harness, rule reference, raw data and analysis scripts, with the defects found during execution and how each was handled.

II. BACKGROUND AND TERMINOLOGY

Structural validation is every check the application performs before business logic: the runtime schema (type, presence, length, pattern, range, enumeration, format and any refinements written into it), service-level checks and database constraints. An input *passes* when the application accepts it. **Semantic validity** is whether a structurally valid value is appropriate for the application’s purpose, covering consistency across fields, domain plausibility, compliance with business rules and safety for downstream consumers such as an LLM component. It is operationalised as satisfying every rule in a frozen, enumerated reference.

An input that passes structural validation and violates at least one rule is **SV-SI** (structurally valid, semantically invalid). The *conditional* SV-SI rate is the share of SV-SI inputs among the structurally valid inputs whose rules were all decided. It is the primary outcome. A generator that produces more well-formed values would otherwise look worse merely by passing more often.

Terminology differs across communities. The OWASP Input Validation Cheat Sheet separates syntactic from semantic validation [22]; I count a range the developer wrote into the schema as structural, because the application enforces it. Security research speaks of logic or business logic vulnerabilities [4], [5], [23], API testing research of inter-parameter dependencies [24], [25]. In LLM security the adjective usually means adversarial intent [26], whereas my usage concerns whether the data is appropriate, which includes that case.

III. RELATED WORK

A. Limits of structural validation, and logic flaws

JSON Schema has a formal semantics and a known complexity [1], [2], and empirical studies describe which constraints practitioners encode [3]. Inter-parameter dependencies such as *requires*, *only-one* and arithmetic relations cannot be stated in an OpenAPI document [24], [25], and schema-derived fuzzers inherit the same boundary [27], and surveys

of input-validation vulnerabilities remain organised around injection classes [28]. The rules beyond that boundary are the classical subject of logic-flaw research. Waler inferred likely invariants over session state and database query parameters and model-checked them [4], and related classes are detected by behavioural patterns [5], parameter tampering [6], mass-assignment testing [7] and excessive-data-exposure fuzzing [29]. Recent work applies LLMs to find such flaws in code [23]. In the studies I reviewed the inputs are crafted by an attacker or a tester, and they are not compared across producers.

B. LLM-generated inputs, and schema-valid but wrong outputs

LLMs are better than random and type-based generators at producing inputs that applications accept. On thirty real forms an LLM-driven approach reaches a form passing rate of 83 %, against 23 % for a static type-based generator and 3 % for random crawling [9], and across 146 forms the best of eleven models submits over 98 % of its inputs [8]. The oracle in both cases is acceptance by the application. Where semantic validity is measured separately, a base model reaches 22.9 % domain-coherent values against 72.4 % after fine-tuning [17]. Related work also mines input constraints from specifications and benchmarks form-filling agents [10]. For model *outputs*, several 2026 studies document the same pattern: OrderBench separates syntactic validity, schema validity and semantic success [11]; hard schema decoding costs accuracy [12]; JSON pass rate stays ahead of value accuracy [13]; schemas shift failures from interface misuse to semantic misuse [14]; and workflows pass validation and then fail to execute [15]. Tool-use benchmarks evaluate emitted calls against a task’s expected answer [30], [31]. In all of them the semantic check is the oracle rather than a layer that is itself evaluated.

C. Data validation beyond schemas, and inappropriate input as a security problem

Data-management research validates data beyond structural schemas with anomaly detection for machine-learning pipelines [32], declarative constraint checks that act as unit tests for data [33] and learned detection of column semantic types [34]; these work on datasets in batch rather than on one request. Classifying a single field entry from a context embedding and statistics of recent entries has been described in a patent [35], of which my EMB design is a simpler, untrained relative. On the security side, indirect prompt injection established that models do not separate data from instructions [36], and that attacks on deployed applications succeed [37]. Prompt fragments inserted through an ordinary form field can later drive a database write [38], the pattern occurs in the wild [39], and benchmarks extend the coverage [40], [41]. Those inputs are attacker-crafted and mostly free text.

D. Detection layers and their cost

Input-side detectors include fine-tuned classifiers [18], embedding classifiers [19], [42], small transformers [43] and

LLM judges [44]. What constrains deployment is the false-positive rate on benign traffic [45]. Cost evidence is thin and tied to GPUs. A detector presented as deployable gives no timing at all [18], an embedding guardrail puts its embedding floor at 35 ms on a datacentre GPU [19], and a layered framework reports a 61 ms median on an A100 with model inference excluded [20]. A systematisation notes that cost and utility are routinely overlooked [21], the one study that compares regex, classifier and LLM tiers head to head does not state its hardware [46], and LLM judges can themselves be redirected by the content they assess [47]. I found no measurement of a validator for data appropriateness on structured request payloads on CPU.

E. Position of this work

Three lines of work come closest among the studies I reviewed, and the design of this study responds to a specific difference in each. The output-side studies measure schema-valid but wrong values that a model emits [11]–[13], [15], where the oracle is the task’s expected answer; I measure values that an application accepts and stores, where it raises no error and the value is later read by people or by an assistant. The form and API testing studies count acceptance by the application [8], [9], so an accepted but inappropriate value counts as a success, and the one study that measures domain coherence separately has no non-LLM comparison [17]; I separate the two outcomes and label semantic validity against a frozen rule reference, independently of the application. The guardrail studies evaluate detectors for adversarial intent and, where they report cost at all, do so on datacentre GPUs [19], [20]; I evaluate detectors for ordinary data appropriateness, on structured payloads, on a four-core CPU, and report detection quality and cost together. The comparison across seven producers, including non-LLM ones, is what lets me ask whether the gap belongs to AI writers or to the validation boundary itself.

IV. RESEARCH QUESTIONS AND HYPOTHESES

The following were frozen, with the primary metrics and the analysis plan, before any experimental input was generated. Prompt family P5 (injection, Section V) is excluded from all primary rates and analysed separately.

- **RQ1.** Do LLM-generated inputs pass structural validation while violating business rules? *H1:* the conditional SV-SI rate of the LLM conditions has a 95 % Wilson interval whose lower bound exceeds 0.05.
- **RQ2.** How does this vary across field categories? *H2:* rates differ across categories; free text and cross-field highest, simple scalar and structured (enumeration) lowest.
- **RQ3.** How do LLM inputs compare with other producers? *H3a:* LLM with field context exceeds random; *H3b:* LLM with field context exceeds benign-unusual reference inputs; *H3c:* LLM without field context exceeds random.
- **RQ4.** Which violation types dominate? Exploratory.

- **RQ5.** Can an added semantic layer detect SV-SI inputs? *H5:* at least one non-judge design reaches recall ≥ 0.70 at a false-positive rate (FPR) ≤ 0.05 on held-out fields.
- **RQ6.** What does it cost? *H6:* rule and embedding designs add at most 50 ms to p95 request latency while an LLM judge adds at least 1000 ms.

V. METHOD

A. Subject under study

The subject is a purpose-built Next.js 16 application written in TypeScript. It uses Zod 4 for validation on the client and the server, Prisma with SQLite for persistence, and a local Ollama runtime for two AI-facing features, support-ticket triage and a booking assistant. It has seven input surfaces and 39 fields. A reference file enumerates 43 business rules, of which five are enforced in code, two of those as Zod refinements (a birth date not in the future, a workshop end after its start). The remaining 38 are unenforced, a profile chosen to resemble small production applications. Each rule is classified in advance as objective, meaning decidable from the input, the database and the clock (19 rules), or as judgement (24 rules). The application and the rule reference were tagged before any input was generated (commit 62a7267), and the commit is recorded with every run. A purpose-built subject is a deliberate trade: real applications do not enumerate their business rules, and without a complete reference the semantic outcome has no ground truth.

B. Targets and generation conditions

Seventeen targets covering 22 fields were selected to represent every field category, every directly AI-facing field, and five cross-field pairs whose rule relates the two members. Each target carries a purpose description, its rule identifiers and its structural constraints. Seven conditions produce values for each target: **A** valid reference values; **B** random values that satisfy the schema, from a seeded generator of word sequences, numbers and dates within the schema’s limits, with no attempt at meaning; **C** rule-based adversarial values, one template per unenforced rule; **D** an LLM given the field name and the structural constraints only; **E** the same LLM given the field’s purpose and rules as well; **F** boundary values at the schema edges; and **G** benign but unusual reference values. Conditions A and G were specified as human-written. In this execution their 272 values were first drafted with the help of a language model, before the main generation, and checked against the schemas and the objective rules. I then went through every value myself, kept 262, replaced 3 with values of my own, deleted 7, and confirmed that the kept values are ones I would have written. The application accepted 0 of the replacements, because all 3 were promotional codes it does not recognise. They are condition A’s only structural rejections, so the reviewed conditions rest on the kept values. I therefore call these conditions human-curated rather than human-written.

C. LLM generation

Conditions D and E use qwen3:4b (Q4_K_M) through Ollama on a four-core CPU, at temperature 0.8, nucleus 0.9, with a fixed seed per run, three runs, and five values per call. Seven prompt families are stored as hashed files: P1 legitimate values, P2 unusual values, P3 values that satisfy the constraints but “would be wrong for the application to accept”, P4 values whose meaning depends on downstream interpretation, P5 injection for AI-facing targets only, P6 boundary values, and P7 Unicode variants. Because P3 asks for violations, results are reported per family as well as pooled. The model identifier, digest, temperature, seed, prompt file and prompt hash are recorded for every value. Failed calls were counted and never regenerated. There were 88 failure records in 52 prompt cells, of which 75 were call errors such as calls aborted at the 120s timeout and 13 were unparseable replies. A further 430 generated values used key names other than the target’s. A harness defect had filled those keys from the form’s default values, so the records measure the defaults rather than the model, and they are treated as generation failures.

D. Structural measurement

Each input is measured twice. It is checked against the same schema object the application imports, and it is submitted over HTTP to the real route handler, which records the status, the service errors and the time. The structural outcome is the application’s own decision, so an input rejected by one of the five enforced business checks counts as a structural rejection. Booking state is reset to the seed before each booking request.

E. Semantic ground truth

Objective rules are decided by rule functions. The 11 objective rules that apply to the 17 targets all have one. Judgement rules are decided by a judge model from a different family than the generator (gemma3:4b, temperature 0). The judge sees the field purpose, the rule text and the submitted value of the target’s fields. It never sees the condition, the prompt family or the intent. The judge does not see the rest of the record. Seven judgement rules depend on other fields, such as capacity being plausible for the location, and those were judged without that context. A sensitivity analysis excludes them. A human annotator labelled 200 structurally valid inputs blind to condition, with 20 items repeated to estimate the annotator’s consistency. Where a human verdict exists it takes precedence over the judge. An input is *decided* when every applicable rule has a verdict. For compute reasons the judge was applied to a stratified sample: 1200 structurally valid inputs drawn round-robin across target and condition with a fixed seed, plus 136 inputs for two booking targets that were re-measured after a defect (Section IX), up to 14 per stratum and as many as were available.

F. Analysis population and statistics

Of the 3881 inputs, the following are excluded: 430 key-mismatch records, 131 records produced by inadvertent retries of failed calls, 74 injection-family inputs, and 10 reference

values that the reviewer deleted or replaced. That leaves 3236 inputs. Of these, 2917 passed structural validation and 1876 are decided. The remaining passes were either not in the judge sample (1001) or received no usable verdict (40). Rates carry Wilson 95 % confidence intervals (CIs), and contrasts are reported as odds ratios (ORs). H1 is evaluated on the lower bound of its interval, H2 with a χ^2 test and Cramér’s V , and H3a to H3c with three one-sided Fisher tests under Holm correction. The judge sample gives the conditions different target mixes, so each contrast is also tested within target with the Cochran–Mantel–Haenszel (CMH) test, which was added after data collection. Values generated in one call, and targets within a category, are not independent, so the intervals are likely too narrow. Per-run and per-target rates are reported for that reason. Every other analysis is labelled exploratory. It was fixed in advance that if judge–human agreement on judgement rules falls below $\kappa = 0.4$, every result that depends on those rules is reported as exploratory. Every number in this paper is generated from the raw data by scripts that are released with it.

VI. SEMANTIC VALIDATION LAYERS

Five designs run as a drop-in check after structural validation. **R** packages the objective rule functions; because those functions also define the ground truth for objective rules, **R** cannot produce a false positive on them and serves as a floor rather than a competitor. **EMB** scores a value by its cosine distance from an embedding (nomic-embed-text) of the field’s purpose and rules, using no labelled examples, unlike the trained classifier the plan anticipated. **SLM** prompts a small language model (SLM; qwen3:1.7b) with the field purpose and rules for a verdict and a confidence. **HYB** runs **R** first and then **EMB** on what **R** passes. **JUDGE** sends the same prompt to qwen3:4b, the generator’s model, and serves as a reference point, so **SLM** and **JUDGE** receive a rubric similar to the ground-truth judge’s. The prompted designs run at temperature 0, and a reply without a verdict counts as a pass, because the deployed layer fails open.

Evaluation is leave-fields-out over five folds. In each fold the threshold is the smallest score at which at most 5 % of the valid training-fold inputs are flagged, and that threshold is applied to the held-out fields. Designs are compared on recall at that operating point, together with FPR, precision and the threshold-free area under the receiver operating characteristic (ROC) curve (AUROC). The model-based designs were scored on a stratified sample of 575 decided inputs, of which 236 are SV-SI, and all designs are compared on that sample. The fast designs were also scored on all 1937 decided inputs, a population that includes the injection family and the retried records.

VII. RESULTS

A. Rate of semantic violations (RQ1)

Among decided, structurally valid LLM inputs, 338 of 1094 (30.9 %, 95 % CI 28.2–33.7) broke at least one business rule (Table I), so H1 is supported. The pooled rate includes prompt

TABLE I

STRUCTURAL AND SEMANTIC OUTCOMES BY INPUT-GENERATION CONDITION. THE CONDITIONAL SV-SI RATE IS THE SHARE OF STRUCTURALLY VALID INPUTS THAT VIOLATE AT LEAST ONE BUSINESS RULE; IT IS THE PRIMARY OUTCOME, COMPUTED OVER *decided* PASSES: THOSE WHOSE EVERY APPLICABLE RULE RECEIVED A VERDICT (PASSES OUTSIDE THE JUDGE SAMPLE ARE UNDECIDED AND EXCLUDED). PROMPT FAMILY P5 (INJECTION) AND RETRIED RECORDS ARE EXCLUDED. INTERVALS ARE WILSON 95% CONFIDENCE INTERVALS. *SPECIFIED AS HUMAN-WRITTEN; VALUES AI-DRAFTED AND THEN REVIEWED VALUE BY VALUE BY THE RESEARCHER (HUMAN-CURATED).

Condition	<i>n</i>	Struct. pass (%)	Passes	Decided	SV-SI	Cond. SV-SI (%)	95% CI
A valid human*	134	97.8	131	130	5	3.8	[1.7, 8.7]
B random	425	93.6	398	362	193	53.3	[48.2, 58.4]
C rule-based adversarial	85	96.5	82	81	56	69.1	[58.4, 78.1]
D LLM context-free	1268	84.0	1065	530	200	37.7	[33.7, 41.9]
E LLM context-aware	1108	92.8	1028	564	138	24.5	[21.1, 28.2]
F boundary	85	96.5	82	81	26	32.1	[22.9, 42.9]
G benign-unusual human*	131	100.0	131	128	22	17.2	[11.6, 24.7]

TABLE II

CONDITIONAL SV-SI RATE OF MODEL-GENERATED INPUTS BY PROMPT FAMILY (DECIDED STRUCTURAL PASSES; P5 EXCLUDED). P3 EXPLICITLY ASKS THE MODEL FOR VALUES THE APPLICATION SHOULD REJECT; P1 ASKS FOR LEGITIMATE VALUES. D: WITHOUT FIELD CONTEXT; E: WITH PURPOSE AND RULES.

Prompt asks for	D (%)	E (%)	SV-SI	D+E (%)	95% CI
P1 legitimate	25.9	14.4	43/212	20.3	[15.4, 26.2]
P2 unusual	36.2	26.1	52/163	31.9	[25.2, 39.4]
P3 rule violations	48.1	39.1	71/164	43.3	[35.9, 50.9]
P4 context-dependent	29.1	20.6	47/193	24.4	[18.8, 30.9]
P6 boundary	46.7	22.3	56/169	33.1	[26.5, 40.5]
P7 Unicode variation	45.6	27.2	69/193	35.8	[29.3, 42.7]
All (H1)			338/1094	30.9	[28.2, 33.7]
All except P3			267/930	28.7	[25.9, 31.7]

family P3, which asks for rule violations. Per family (Table II), the rate was 20.3 % for legitimate-value prompts and 43.3 % when violations were requested. Without P3 it was 28.7 %. Agreement between the judge and the human annotator is below the pre-registered threshold (Section VII E), so every rate that depends on judgement rules is exploratory, this one included. The objective-rule rate below depends only on the rule functions and is not affected. H1 holds under every sensitivity variant in Table III, including objective rules alone (7.9 %, CI 6.8–9.2), which do not depend on the judge. One target dominates: on workshop dates (F15+F16) nearly every model value was in the past, because the prompts never state the current date, and without that target the rate is 21.5 %. Per run the rate was 31.1, 41.6 and 40.9 % without field context and 25.7, 24.9 and 22.9 % with it; including the retried records gives 30.5 %. The reference conditions bound the labelling from both sides: valid inputs were labelled invalid in 3.8 % of cases and adversarial inputs in 69.1 %, so about a third of the designed violations were missed and a few valid inputs were flagged.

B. Field categories (RQ2)

Category is associated with the outcome ($\chi^2(6) = 184.5$, $p < 0.001$, Cramér’s $V = 0.31$, a moderate association), so H2’s test is supported (Table IV). These rates rest on judgement rules as well, so they are exploratory. The directional

TABLE III

SENSITIVITY OF THE CONDITIONAL SV-SI RATE TO DESIGN CHOICES. LLM: CONDITIONS D AND E POOLED; B: RANDOM; G: BENIGN-UNUSUAL (HUMAN-CURATED). “WITHOUT CONTEXT RULES” DROPS SEVEN JUDGEMENT RULES THE JUDGE HAD TO DECIDE WITHOUT THE REST OF THE RECORD. “OBJECTIVE RULES ONLY” USES NO JUDGE. TARGET-WEIGHTED AVERAGES PER-TARGET RATES BY EACH TARGET’S SHARE OF STRUCTURAL PASSES. P1 AND P3 APPLY TO MODEL CONDITIONS ONLY.

Variant	LLM (%)	95% CI	B (%)	G (%)
Pre-registered (all rules)	30.9	[28.2, 33.7]	53.3	17.2
P1 (legitimate) prompts only	20.3	[15.4, 26.2]	–	–
Without P3 (asks for violations)	28.7	[25.9, 31.7]	53.3	17.2
Without dates (F15+F16)	21.5	[19.0, 24.2]	50.4	17.5
Objective rules only	7.9	[6.8, 9.2]	10.3	0.8
Without context rules	24.5	[22.1, 27.2]	36.5	9.4
Target-weighted	30.2	–	53.3	17.2

TABLE IV

CONDITIONAL SV-SI RATE BY INPUT-FIELD CATEGORY, OVER DECIDED STRUCTURAL PASSES FROM ALL CONDITIONS. A χ^2 TEST OF INDEPENDENCE GIVES $\chi^2 = 184.5$, $p < 0.001$, Cramér’s $V = 0.314$.

Field category	Decided	SV-SI	Cond. SV-SI (%)	95% CI
Structured	107	61	57.0	[47.5, 66.0]
Free text	69	35	50.7	[39.2, 62.2]
Cross-field	600	270	45.0	[41.1, 49.0]
AI-facing	260	114	43.8	[37.9, 49.9]
Search	95	30	31.6	[23.1, 41.5]
Simple scalar	249	68	27.3	[22.2, 33.2]
Financial	496	62	12.5	[9.9, 15.7]

expectation is only partly met. Free text (51 %) and cross-field (45 %) are high, as expected. Structured fields are the highest (57 %) rather than among the lowest, and financial fields are the lowest (12 %). Categories contain between one and four targets, and per-target rates are in the repository. The structured category is a single target, a ticket’s priority and category, whose rules the judge decided without seeing the ticket text, so the reversal is not a property of enumerated fields in general.

C. Comparison of producers (RQ3)

None of the three pre-registered contrasts is supported (Table V), and two point the other way. They are computed

TABLE V

PRE-DECLARED ONE-SIDED CONTRASTS ON THE CONDITIONAL SV-SI RATE (FISHER EXACT, HOLM-CORRECTED ACROSS THE THREE TESTS). EACH TEST IS ONE-SIDED IN THE PRE-REGISTERED DIRECTION (RATE 1 > RATE 2), SO AN ODDS RATIO BELOW 1 YIELDS p CLOSE TO 1. CONDITIONS: B RANDOM, E LLM WITH FIELD CONTEXT, G BENIGN-UNUSUAL (HUMAN-CURATED), D LLM WITHOUT FIELD CONTEXT. WITHIN-TARGET RESULTS ARE IN THE TEXT.

	Contrast	Rate 1 (%)	Rate 2 (%)	OR	p_{Holm}	Supported
H3a	E>B	24.5	53.3	0.28	1.00	no
H3b	E>G	24.5	17.2	1.56	0.14	no
H3c	D>B	37.7	53.3	0.53	1.00	no

on labels that include judgement rules, so under the agreement gate they are exploratory as well. The objective-rule sensitivity row shows the same direction without the judge. Random structurally valid inputs broke rules in 53.3 % of cases, more often than LLM inputs without field context (37.7 %) or with it (24.5 %), and that holds in every sensitivity variant (Table III). Within target (CMH, post hoc) the odds of a violation were lower for LLM inputs with context than for random inputs (odds ratio 0.16, 95 % CI 0.11–0.23) and lower without context (0.47, 0.34–0.66); adversarial inputs exceeded random ones (2.18, 1.20–3.95), and field context lowered the odds relative to the schema alone (0.23, 0.15–0.35). LLM inputs with context and benign-unusual reference inputs did not differ detectably (1.32, 0.77–2.27, $p = 0.29$); the interval is wide, and the reference inputs are AI-drafted values curated by a human reviewer, so this compares model input with human-curated input rather than with input a person wrote unaided.

An exploratory variation repeated conditions D and E on ten targets with families P1 to P4, changing only the generator. The 1.7-billion-parameter model produced more violations than the 4-billion model, both on objective rules (4.8 % against 1.1 %, odds ratio 4.4, $p_{\text{Holm}} = 0.001$) and on all rules (35.0 % against 22.3 %, $p_{\text{Holm}} = 0.02$). Neither the 8-billion model (23.1 %) nor temperature 0 (26.3 %) differed detectably. With one run per generator and a smaller judge sample, this only indicates that a smaller model may widen the gap.

D. Violation types (RQ4)

Counts of judgement-rule violations depend on the judge and are exploratory. Judgement-rule violations outnumber objective ones (502 against 272, across all conditions). The single most frequent violation is objective: a workshop must start in the future (146). It is followed by a profile bio addressed to the assistant (64), a ticket priority inconsistent with its text (61) and duration plausibility (57). The ticket rule was judged without the text it depends on. On AI-facing fields, 81 structurally valid LLM inputs violated a rule. Of those, 20 came from the injection family and 61 from the other prompt families.

E. Ground-truth agreement

On judgement rules, the judge agreed with the human annotator on 70.6 % of 126 rule verdicts where both gave

a verdict (Cohen’s $\kappa = 0.24$); the annotator was unsure on 3.6 % of judgement-rule verdicts. Agreement is below the pre-registered 0.4, so every judgement-rule result in this paper is exploratory. The rule functions agreed with the annotator on 89.5 % of objective-rule verdicts ($\kappa = 0.32$), and the annotator’s repeated annotations agreed with each other on 93.8 % ($\kappa = 0.71$). The objective-rule results below are decided by those rule functions rather than by the judge, but that single human check of them is itself weak, so they are better described as independent of the judge than as externally validated.

F. Detection quality (RQ5)

No design reached the target, so H5 is not supported (Table VI). Detection is measured against the same labels, so the judgement-rule part of these results is exploratory as well. The objective-rule part is not. At the operating point fixed on the training folds, the best non-judge design, HYB, detected 27.1 % of SV-SI inputs (CI 21.3–32.5) at a held-out FPR of 5.6 %, which is above the 5 % it was tuned for. Most of what it catches comes from R. HYB and R both caught 100.0 % of the inputs that violate only objective rules, and 6.5 % and 0.0 % of those that violate a judgement rule. On all decided inputs the fast designs reach 33.3 % for R and 36.6 % for HYB. EMB on its own was no better than chance (AUROC 0.45).

The prompted designs rank inputs better (AUROC 0.75 for SLM and 0.76 for JUDGE against 0.60 for HYB), but their scores take only a handful of values, so holding FPR to 5 % on the training fields forces the threshold to the top confidence level, where SLM detects 1.3 % and JUDGE 23.7 %. Taking each model’s own verdict without tuning, which is exploratory, SLM detects 69.1 % while flagging 20.1 % of valid inputs, and JUDGE detects 51.7 % while flagging 6.8 % and gives no usable verdict for 26 % of inputs. Both receive a rubric similar to the ground-truth judge’s, so their agreement with it partly measures agreement between two LLMs. Thresholds did not transfer across fields: the held-out FPR per fold reached 30 % for the embedding-based designs. On benign-unusual inputs the FPR was 4.8 % for HYB, 1.6 % for SLM and 1.6 % for JUDGE.

G. Overhead (RQ6)

Each configuration ran against the production build with only its own model loaded. No request failed and no layer call failed in any run (Table VII). H6 is not supported, because its first part fails. R added at most 1 ms to p97.5 latency with one client and 14 ms with eight, which is within the 50 ms bound. The negative differences at eight clients reflect run-to-run variation. EMB added 65–96 ms with one client, above the bound, and 122–751 ms with eight, because every request embeds its text on the same CPU. On the p95 bracket used as a sensitivity check, EMB fails on the review endpoint and is undetermined on the other two. The second part of H6 holds. SLM added 527–534 ms at the median with one client and JUDGE added 1611–7704 ms. The model server answers one request at a time, so requests queue under concurrency. With

TABLE VI

DETECTION QUALITY OF THE CANDIDATE SEMANTIC VALIDATION LAYERS ON THE SHARED STRATIFIED SAMPLE OF 575 DECIDED INPUTS (236 SV-SI, 339 VALID), LEAVE-FIELDS-OUT (5 FOLDS). RECALL, 95% BOOTSTRAP INTERVAL, FPR AND PRECISION ARE AT THE PRE-REGISTERED OPERATING POINT: THE THRESHOLD IS THE SMALLEST SCORE WHOSE FALSE-POSITIVE RATE ON VALID TRAINING-FOLD INPUTS IS AT MOST 5%, APPLIED TO HELD-OUT FIELDS. AUROC IS THRESHOLD-FREE. THE NATIVE COLUMN (EXPLORATORY, NOT PRE-REGISTERED) APPLIES EACH DESIGN’S OWN DECISION RULE WITHOUT TUNING: FOR SLM AND JUDGE, THE MODEL ANSWERED “NOT APPROPRIATE”; FOR R, EMB AND HYB, SCORE ≥ 0.5 . JUDGE RETURNED NO USABLE VERDICT FOR 26% OF INPUTS, COUNTED AS NOT FLAGGED. R’S ZERO FPR IS PARTLY BY CONSTRUCTION: ITS RULE FUNCTIONS ALSO DECIDE THE GROUND TRUTH FOR OBJECTIVE RULES. JUDGE SHARES A MODEL FAMILY WITH THE GENERATOR AND IS A REFERENCE POINT. LATENCY IS THE MEDIAN DECISION TIME OF THE LAYER ALONE ON THE STUDY CPU; HYB’S FIGURE REUSES EMBEDDINGS ALREADY CACHED BY EMB, SO ITS UNCACHED REQUEST-PATH COST IS THE ONE IN TABLE VII.

Design	Recall (%)	95% CI	FPR (%)	Precision (%)	AUROC	Native recall / FPR (%)	Median (ms)
HYB	27.1	[21.3, 32.5]	5.6	77.1	0.60	24.2 / 2.1	0.01
JUDGE	23.7	[18.5, 29.4]	3.5	82.4	0.76	51.7 / 6.8	2709
R	22.5	[17.0, 27.8]	0.0	100.0	0.61	22.5 / 0.0	0.01
EMB	4.7	[2.2, 7.3]	5.6	36.7	0.45	1.7 / 2.1	31
SLM	1.3	[0.0, 2.9]	1.2	42.9	0.75	69.1 / 20.1	1075

TABLE VII

REQUEST-PATH OVERHEAD OF EACH CONFIGURATION (PRODUCTION BUILD, FOUR-CORE CPU, NO GPU). EACH VALUE IS THE MEDIAN OVER THREE 60 s RUNS, GIVEN AS THE RANGE ACROSS THE THREE ENDPOINTS (PROFILE, REVIEW, COURSE). ADDED LATENCY IS RELATIVE TO STRUCTURAL-ONLY VALIDATION ON THE SAME ENDPOINT AND CONCURRENCY. H6 IS EVALUATED ON THE ADDED p97.5, THE NEAREST TAIL PERCENTILE THE LOAD GENERATOR REPORTS. “OK” IS THE SHARE OF LAYER CALLS THAT RETURNED A VERDICT RATHER THAN FAILING OPEN. “>60 000”: THE CONFIGURATION SATURATED, COMPLETING NO REQUEST WITHIN AT LEAST TWO OF THE THREE RUNS. PER-ENDPOINT VALUES ARE IN THE REPOSITORY.

Config.	$\Delta p50$ (ms)	$\Delta p97.5$ (ms)	req/s	Ok (%)	H6
<i>Concurrency 1</i>					
structural	0	0	42.9–47.4	–	–
R	0–2	0–1	41.9–46.0	100	holds
EMB	27–48	65–96	13.2–19.2	100	fails
HYB	26–45	65–82	13.6–19.2	100	–
SLM	527–534	655–761	1.7–1.8	100	–
JUDGE	1611–7704	7683–10824	0.1–0.5	100	holds
<i>Concurrency 8</i>					
structural	0	0	41.7–48.3	–	–
R	2–6	–136–14	43.9–47.6	100	holds
EMB	126–288	122–751	16.0–27.9	100	fails
HYB	103–270	131–396	17.7–29.0	100	–
SLM	4149–4210	6780–7180	1.7–1.7	100	–
JUDGE	14282–>60 000	23474–>60 000	0.0–0.4	100	holds

eight clients SLM added 4149–4210 ms at the median, and JUDGE completed no request within the 60 s window in all three runs on the course endpoint and in two of three on the review endpoint. Throughput on the course endpoint fell from 47.4 requests per second with structural validation alone to 1.8 with SLM and 0.1 with JUDGE. Set against detection quality (Fig. 1), only R and HYB lie on the trade-off frontier. SLM, JUDGE and EMB each cost more and detect less than one of them at the pre-registered operating point.

VIII. DISCUSSION

Throughout this section, every statement that rests on judgement rules is exploratory, because judge–human agreement fell below the pre-registered threshold. Statements about objective rules do not depend on the judge.

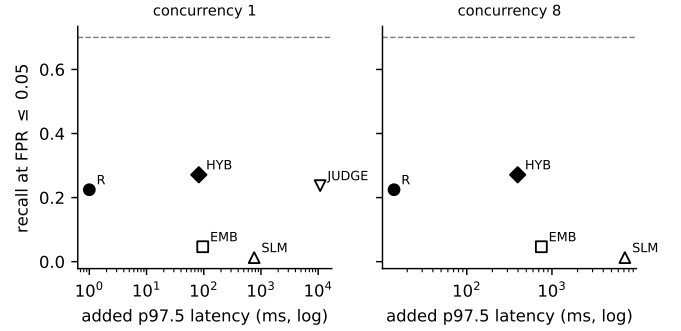


Fig. 1. Detection versus request-path cost for each semantic layer: recall at $FPR \leq 0.05$ on held-out fields (Table VI) against the largest added p97.5 latency over the three endpoints (Table VII), per concurrency level. Filled markers lie on the trade-off frontier. The dashed line is the pre-registered 70% recall target. JUDGE at concurrency 8 saturated and is drawn at its lower bound.

RQ1: the gap is measurable, and its size depends on what is asked. About a third of structurally valid LLM inputs broke an unenforced rule. The figure is a fifth when the model was asked only for legitimate values, and under a tenth on objective rules alone. This extends output-side findings [11], [13], [15] to inputs that an application would store. The magnitude is sensitive to prompt intent, to one date-dependent target, and to the judge, whose agreement with a human annotator was $\kappa = 0.24$. The claim that survives every check is that the gap is above the 5% bound, not any single figure.

RQ2: where it occurs is uneven. Category is moderately associated with the outcome, but individual targets drive that association. One target with a clock-dependent rule accounts for over half of the cross-field violations, and the structured category is a single target that the judge scored without its context.

RQ3: the gap is not specific to AI. This is the central negative result. Random values that satisfy the schema broke rules more often than model values, with or without field context, both pooled and within target. The results point to the validation boundary rather than to the producer. A schema

admits many values the application should reject, and any producer that satisfies the schema without knowing the rules will supply some of them. What distinguishes the model is that context helps. Giving it the purpose and the rules cut the odds of a violation by 77 % relative to the schema alone. The random generator produces word sequences with no attempt at meaning, so it is a weak producer, and the comparison bounds non-AI input rather than characterising it.

RQ4: most violations are ordinary. The single most frequent violation was of an objective rule that the application could enforce directly, so enforcing objective rules would remove part of the gap. Judgement-rule violations are still the majority. On AI-facing fields most LLM violations came from ordinary prompts rather than from injection attempts, so the gap is mostly unenforced business logic rather than attack.

RQ5: no cheap layer closes it. The fast designs that can be held to a low FPR, R and HYB, find about a quarter of the violations at the operating point, and most of what they find are objective violations that could simply be enforced. JUDGE at a similar FPR also finds about a quarter, split evenly between objective and judgement rules. The prompted models discriminate better but produce coarse, field-dependent scores. Used as they are, they catch more while rejecting between one valid input in fifteen and one in five in this sample. Detecting judgement-rule violations remains the open problem.

RQ6: cost rises faster than detection. On commodity CPU hardware only the rule layer is effectively free, and it detects violations of rules the application could have enforced in the first place. The embedding layer costs tens of milliseconds per request and detected almost nothing here. The prompted models cost half a second to several seconds per request with one client, a local model server handles one request at a time, so they cannot keep up with eight, and their better ranking of inputs does not survive the false-positive constraint. On this hardware the practical recommendation is to enforce objective rules in code and to treat judgement rules as an open problem, rather than to add a model to the request path.

Alternative explanations. The six explanations fixed in advance fare as follows. *Weak business rules rather than AI:* consistent with the data, because the gap appears for every producer that does not know the rules. *Random inputs produce the same effect:* consistent, and random inputs exceed model inputs. *Unusual human inputs behave the same:* not resolved. Human-curated unusual inputs did not differ detectably from context-aware model inputs (1.32, 0.77–2.27), but the interval spans both directions and the contrast is exploratory. *Confined to free text:* not supported, because structured and cross-field targets also show high rates, although single targets drive them. *Prompt injection under another name:* not supported. Injection prompts are excluded from all rates and account for 20 of 81 LLM violations on AI-facing fields. *Too expensive to deploy:* supported for the model-based layers on CPU hardware, and not supported for rules.

IX. THREATS TO VALIDITY

Internal validity. Labelling is the main threat. Judgement rules were decided by one 4B judge (agreement with a human annotator $\kappa = 0.24$ on 126 shared verdicts, below the pre-registered 0.4, so judgement-rule results are exploratory), and seven of them were decided without the fields they depend on. Excluding those seven lowers the LLM rate to 24.5 % and lowers the random rate by more (Table III), which leaves the RQ3 direction intact. Agreement on semantic judgements can be low even between humans [48]. The disagreement here is not one-sided noise. On the shared verdicts the judge called 37 violations against the annotator’s 28, so the judge is the stricter rater and judgement-rule rates may be too high. The judge also returned no usable verdict on 36 further rules that the annotator decided, and the agreement statistic cannot use those. The annotator in turn disagreed with the deterministic rule functions on objective rules ($\kappa = 0.32$, 89.5 % agreement, 19 rule-function violations against 4), and those functions decide such rules from the record and the clock. Part of the low agreement is therefore annotation error, or the limited context the annotator was shown, rather than judge error alone. The annotator was self-consistent on the repeated items ($\kappa = 0.71$). The judge sample gives the conditions different target mixes, and target-weighted rates are close to the pooled ones. Values generated in one call are correlated, so the intervals understate uncertainty. Time-relative rules are evaluated against the clock at the moment the labels are merged, so a later re-run can change labels near the boundary. The judge ran once, and its run-to-run stability on this CPU was not measured.

Defects found and corrected. Six defects were found during execution, each recorded as a deviation with the invalid data preserved: the database was not reset between booking measurements, so capacity ran out and those targets were re-measured; a resume routine retried failed generation calls, whose records are excluded; model outputs with other key names were silently replaced by form defaults, and those records are excluded as generation failures; a first overhead run never enabled the layer and is discarded; the first detector analysis chose thresholds with a quantile, which fails for coarse scores, and was replaced before any detector result was reported; and the label merge never passed human verdicts to the ground-truth algorithm while the importer dropped cells after an empty one, both fixed before the annotation was collected. Three of the six were found by independent review of the analysis and the code.

External validity. The study covers one application in one domain, built by the researcher, with one enforcement profile, so the results may not transfer. Generation uses quantised open models of 1.7 to 8 billion parameters on CPU. Frontier models may behave differently, and the evidence that scale does not predict semantic quality [13], [15] cuts both ways. The prompts never state the current date, which inflates date-rule violations for model inputs.

Construct validity. Semantic invalidity is defined by a frozen reference that another team would write differently. The

reference was frozen before any input existed, and violations are reported per rule, so a reader can discount rules they disagree with.

Conclusion validity. Pooling three runs inflates the LLM sample, and per-run rates are reported for that reason. The category mapping assigns dual-tagged fields to one category, which was decided before the data existed but not logged at the time. The three contrasts H3a to H3c were the only pre-declared ones, but they are computed on labels that include judgement rules, so they too are exploratory under the pre-registered agreement gate, and only results decided by the rule functions alone remain confirmatory.

Who produced the human data. I carried out both the review and the annotation, and I also built the subject application and wrote the rule reference. The annotation is blind to the generation condition, because the interface never shows it, but it is not independent of the study, and a single annotator cannot establish how far the disagreement with the judge generalises. The review took 19 minutes for 272 values, a median of 1 s per value, and the annotation took 33 minutes for 200 items, a median of 3 s per item. Both are well below the effort budgeted for them, which may itself contribute to the low agreement. The per-item answers and their timestamps are released with the artefacts.

Provenance. The reference conditions A and G were specified as human-written and their first drafts came from a language model. I then reviewed and curated every value, so H3b compares model input with human-curated input rather than with input written unaided. H3b remains exploratory because it depends on judgement rules.

X. CONCLUSION AND FUTURE WORK

Schema validation answers whether an input is well formed, not whether it belongs. At the input boundary of one application with an explicit rule reference, about a third of structurally valid LLM inputs broke an unenforced business rule. Random structurally valid inputs broke rules more often, and telling the model the field's purpose and rules reduced violations. The gap is therefore a property of what the application leaves unenforced and is not specific to AI-generated input. The single most frequent violation was of an objective rule that a line of validation code could enforce. None of the five candidate layers detected the pre-registered share of violations at a tolerable false-positive rate. A human annotation of 200 inputs agreed with the judge only at $\kappa = 0.24$, below the threshold fixed in advance, so every result that depends on judgement rules is exploratory, and the ground truth for those rules is the weakest part of the study. On a four-core CPU, rules added at most 14 ms at p97.5 and embeddings tens of milliseconds. At the median a small language model added about half a second and an LLM judge several seconds, and the judge saturated on two of three endpoints with eight concurrent clients.

Future work should strengthen the ground truth with several annotators and more time per item, collect reference

inputs written unaided by several people, re-judge the context-dependent rules with the full record, and repeat the protocol on an existing open-source application.

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